

# GAP ANALYSIS REPORT

## E-Commerce Order Fulfillment Process

*As-Is to To-Be Assessment for a Mid-Sized Indian E-Commerce Company*

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### 1. Executive Summary

This Gap Analysis Report examines the current order fulfillment process of a mid-sized Indian e-commerce company operating a marketplace model (comparable to Flipkart or Meesho) that is experiencing systemic delivery delays. The report identifies ten critical process areas, documents the performance gap between the current and desired state, and prescribes a prioritised set of interventions to bring the organisation to best-in-class fulfillment performance.

#### Key findings at a glance:

- Average order-to-delivery cycle time: 4–7 days (industry benchmark: 1–2 days for metro, 3–4 days for Tier 2/3)
- First-attempt delivery failure rate: ~20% (industry best practice: <10%)
- Seller acceptance delay contributes 30–40% of total fulfillment time
- WISMO (Where Is My Order) tickets account for ~30% of all customer care volume
- Return processing takes 7–10 days — well above the 2–3 day benchmark
- 10 process areas assessed; 4 rated High Impact, 4 Medium Impact, 2 Low Impact

### 2. Methodology

The analysis was conducted using the following approach:

- Process walkthroughs with Operations, Technology, Logistics, Seller Management, and Customer Care teams
- Analysis of OMS, WMS, and logistics platform data over a 90-day period
- Benchmarking against industry standards and comparable Indian e-commerce operators
- Root cause analysis (5-Why, fishbone) applied to the top three delay categories
- Stakeholder workshops to validate findings and co-create desired-state definitions

Each process area was assessed across five dimensions: cycle time, error rate, system integration maturity, customer experience impact, and seller experience impact.

### 3. Current State Summary (As-Is)

The current fulfillment process spans six phases: Order & Payment, Seller Acceptance & Packing, First-Mile Pickup, Sorting Hub Operations, Last-Mile Delivery, and Post-Delivery Settlement. The end-to-end process involves the Customer, OMS, Seller, multiple Logistics Partners, and the Finance team.

#### 3.1 Key Metrics — Current State

|                                 |                                               |
|---------------------------------|-----------------------------------------------|
| Order confirmation time         | 15–30 minutes (target: <2 min)                |
| Seller acceptance time          | Avg. 45 minutes; 15% rejection rate           |
| Pick & pack time                | 45–135 minutes per order                      |
| First-mile pickup lead time     | 2–8 hours from scheduling; 10% no-show rate   |
| Hub misrouting rate             | 3–4% of packages; adds 24–48 hrs per incident |
| Intercity transit time          | 6–72 hours (truck-dependent)                  |
| Last-mile first-attempt success | ~80% (20% failure rate)                       |
| Return-to-Origin (RTO) rate     | ~8% of all orders                             |
| Return pickup SLA               | 5–7 business days                             |
| Refund processing time          | 7–10 days post return receipt                 |
| Seller settlement cycle         | T+7 to T+14 days                              |
| WISMO customer care tickets     | ~30% of total contact volume                  |

#### 3.2 Root Causes of Delivery Delays

Structured analysis identified four systemic root causes:

- Fragmented technology stack: OMS, WMS, logistics platforms, and seller dashboards are not integrated in real-time, creating information gaps that cause manual intervention at every handoff.
- Seller ecosystem immaturity: A significant portion of sellers operate without ERP systems, rely on manual stock counts, and are not incentivised by SLA-linked performance models.
- Last-mile execution gaps: DE load balancing is manual, customer availability is not pre-confirmed, and the DE mobile app fails under low-connectivity conditions common in Tier 2/3 geographies.
- Reactive operations: Decisions at hubs and dispatch centres are based on batched, overnight data — no real-time alerting system exists to flag delays before they compound.

## 4. Desired Future State (To-Be)

The desired future state defines a fully orchestrated, event-driven fulfillment engine where every step from cart to delivery is tracked in real time, exceptions are handled automatically, and both seller and customer experience is measurably improved.

### 4.1 Key Metrics — Desired State

|                                        |                                                         |
|----------------------------------------|---------------------------------------------------------|
| <b>Order confirmation time</b>         | <2 minutes (fully automated)                            |
| <b>Seller acceptance time</b>          | <5 minutes for gold-tier (auto-accept); <15 min overall |
| <b>Seller rejection rate</b>           | <3% (real-time stock sync eliminates mismatch)          |
| <b>Pick &amp; pack time</b>            | <30 minutes (WMS-guided scanning)                       |
| <b>First-mile pickup lead time</b>     | <4 hours; <2% no-show rate                              |
| <b>Hub misrouting rate</b>             | <0.5% (IoT-enhanced sorter + QC scan)                   |
| <b>Intercity transit time</b>          | Optimised via dynamic truck dispatch                    |
| <b>Last-mile first-attempt success</b> | >85%                                                    |
| <b>Return-to-Origin (RTO) rate</b>     | <3%                                                     |
| <b>Return pickup SLA</b>               | Within 48 hours                                         |
| <b>Refund processing time</b>          | Initiated on pickup scan; credited within 2 days        |
| <b>Seller settlement cycle</b>         | T+2 for verified sellers (event-triggered)              |
| <b>WISMO customer care tickets</b>     | <5% of total contact volume                             |

### 4.2 Enabling Capabilities

- Unified logistics orchestration API — single integration point for all carrier partners, with AI-based carrier selection per pin code and order type
- Real-time event streaming — Kafka/SQS-based event bus connecting OMS, WMS, logistics, and customer notification systems
- Seller Intelligence Platform — SLA scoring, auto-accept rules, stock API mandates, and tiered incentive structures
- AI-powered last-mile routing — dynamic DE route optimisation accounting for package weight, time windows, and historical delivery patterns
- Offline-first DE mobile app — all critical actions (scan, OTP, photo) work offline and sync on connectivity restore
- Self-serve customer return portal — eligibility check, label generation, and real-time return status in-app

## 5. Gap Analysis Table

The table on the following pages presents the detailed gap analysis across ten process areas. Each row includes the current state, desired state, identified gap, business impact rating, recommended solution, and implementation priority.

### Impact Rating Key:

- High — Direct, immediate impact on delivery SLA, customer satisfaction, or revenue
- Medium — Significant but secondary impact; worsens experience if unresolved
- Low — Operational efficiency or long-term strategic benefit

### Priority Key:

- P1 – Critical: Must be initiated within 0–3 months; blocking other improvements
- P2 – High: 3–9 months; significant ROI, non-blocking
- P3 – Medium: 9–18 months; strategic or analytical improvements

| Process Area                                   | Current State (As-Is)                                                                                                   | Desired State (To-Be)                                                                               | Gap Identified                                                                                                                                | Impact | Recommended Solution                                                                                                                                                       | Priority      |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>1. Order Processing &amp; Payment</b>       | Manual OMS; 8-12% payment failure rate; no real-time inventory sync; avg 15-30 min to confirm order                     | Automated OMS with real-time inventory; smart payment retry; order confirmed in <2 min              | <i>No payment retry intelligence; OMS not integrated with seller ERP; phantom inventory causes ~5% cancellations</i>                          | High   | Implement intelligent payment retry engine; integrate OMS with seller ERP via API; adopt real-time inventory reservation on cart add                                       | P1 – Critical |
| <b>2. Seller Onboarding &amp; Acceptance</b>   | Avg. 45 min seller response time; 15% rejection rate due to stock mismatch; no auto-escalation; manual seller dashboard | Auto-acceptance for verified sellers; <5 min confirmation; rejection rate <3%; real-time stock sync | <i>No automated acceptance rules; sellers not incentivised for fast response; stock data stale by 6-12 hrs</i>                                | High   | Build seller SLA scoring system; introduce auto-accept for gold-tier sellers; enforce hourly stock feed via seller API; add penalty/reward model                           | P1 – Critical |
| <b>3. Warehouse &amp; Inventory Management</b> | Manual pick-list; no slotting optimisation; 3-5% pick errors; avg 45-135 min pick-pack per order                        | WMS-driven pick lists with zone routing; pick error <0.5%; avg pick-pack <30 min                    | <i>No WMS slotting intelligence; paper-based pick lists; no cycle count programme; dead stock blocks bin locations</i>                        | High   | Deploy WMS with velocity-based slotting; introduce handheld scanner picking; run weekly cycle counts; implement FIFO bin rules                                             | P1 – Critical |
| <b>4. First-Mile Logistics &amp; Pickup</b>    | Multiple logistics portals; 10% agent no-show rate; manual manifesting; rejected packages add 24-48 hrs                 | Unified logistics API; <2% no-show; digital manifest; same-day re-pickup on rejection               | <i>No single logistics orchestration layer; no real-time agent tracking; package rejection SLA undefined; no automated rescheduling</i>       | High   | Build logistics orchestration middleware (single API, multi-carrier routing); add agent GPS tracking; set 4-hr re-pickup SLA; auto-notify seller on rejection              | P1 – Critical |
| <b>5. Sorting Hub Operations</b>               | 3-4% misrouting rate; sorter downtime during peak; truck consolidation adds 2-12 hr wait; no live tracking between hubs | Misrouting <0.5%; 24/7 sorter uptime; dynamic truck dispatch; end-to-end hub tracking               | <i>Aging conveyor tech; no predictive maintenance; static truck schedules; hub-to-hub tracking not integrated into customer-facing status</i> | High   | Install IoT sensors on sorters for predictive maintenance; implement dynamic truck dispatch triggered by volume thresholds; integrate hub scans into customer tracking API | P2 – High     |
| <b>6. Last-Mile Delivery</b>                   | 20% first-attempt delivery failure;                                                                                     | First-attempt success >85%; AI-optimised                                                            | <i>No intelligent time-slot</i>                                                                                                               | High   | Introduce delivery time-slot selection at checkout; deploy AI route optimiser;                                                                                             | P1 – Critical |

| Process Area                                     | Current State (As-Is)                                                                                                  | Desired State (To-Be)                                                                              | Gap Identified                                                                                                                                                         | Impact | Recommended Solution                                                                                                                                                                       | Priority           |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|                                                  | manual DE assignment; app crashes in low connectivity; RTO rate ~8%                                                    | route assignment; offline-capable DE app; RTO <3%                                                  | <i>pre-selection; load balancing is manual; DE app not offline-first; no proactive customer availability check</i>                                                     |        | rebuild DE app with offline sync; add SMS/WhatsApp confirmation 1 hr before delivery                                                                                                       |                    |
| <b>7. Customer Communication &amp; Tracking</b>  | SMS confirmation only; 5-30 min status lag; no live GPS tracking; 30% of customer care tickets are 'where is my order' | Omnichannel updates (SMS + WhatsApp + email + app); <1 min status lag; live GPS; WISMO tickets <5% | <i>No unified communication orchestration; status events not streamed in real-time; no customer-facing live map; care team cannot see same data as customer</i>        | Medium | Implement event-driven notification engine (Kafka/SQS); expose tracking API to customer app with live DE GPS; unify care dashboard with same real-time feed                                | <b>P2 – High</b>   |
| <b>8. Returns &amp; Reverse Logistics</b>        | Manual return initiation; 5-7 day pickup SLA; no quality check at doorstep; refund takes 7-10 days post-return receipt | Self-serve return in app; 48-hr pickup; doorstep QC; refund initiated on pickup scan               | <i>No automated return eligibility engine; reverse logistics not integrated with forward logistics fleet; quality assessment is manual and slow</i>                    | Medium | Build return eligibility API with policy rules engine; integrate reverse pickup into DE daily route; introduce doorstep photo QC on DE app; trigger instant refund on scan                 | <b>P2 – High</b>   |
| <b>9. Seller Payment Settlement</b>              | T+7 to T+14 settlement cycle; manual dispute resolution; no self-serve settlement dashboard for sellers                | T+2 settlement for verified sellers; automated dispute triage; real-time settlement dashboard      | <i>Payment engine not event-driven; disputes handled via email; sellers lack visibility into deductions; cash-flow strain increases rejection rate (feedback loop)</i> | Medium | Move to event-driven settlement triggered by delivery confirmation; build seller portal with deduction breakdown; introduce BNPL/advance payout option for high-performing sellers         | <b>P2 – High</b>   |
| <b>10. Analytics &amp; Decision Intelligence</b> | Batch reports run nightly; ops decisions based on yesterday's data; no predictive demand or delivery forecasting       | Real-time operational dashboards; predictive delay alerts; ML-based demand forecasting             | <i>No streaming data pipeline; data siloed across OMS, WMS, logistics, and CRM; no ML model in production</i>                                                          | Low    | Build unified data lake with Kafka-based streaming ingestion; deploy real-time ops dashboard (Grafana/Superset); pilot ML delay prediction model; run demand forecasting for top 1000 SKUs | <b>P3 – Medium</b> |

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## 6. Implementation Roadmap

The recommendations are sequenced across three phases to balance speed-to-impact with change management capacity. Phase 1 focuses on high-visibility quick wins that will reduce delays within 90 days. Phase 2 delivers the core technology transformation. Phase 3 builds the analytics and intelligence layer.

| Phase 1: Quick Wins (0–3 months)         | Phase 2: Core Transformation (3–9 months)   | Phase 3: Optimisation (9–18 months)                 |
|------------------------------------------|---------------------------------------------|-----------------------------------------------------|
| Intelligent payment retry engine         | Logistics orchestration API layer           | Streaming analytics platform & real-time dashboards |
| Seller auto-accept rules + SLA scoring   | AI route optimiser for last-mile DEs        | ML-based demand & delay forecasting                 |
| Omnichannel customer notification engine | WMS with velocity-based slotting + scanners | Full reverse logistics integration                  |
| Doorstep QC on DE app for returns        | Event-driven seller settlement (T+2)        | Predictive sorter maintenance (IoT)                 |
| Seller stock API (hourly feed)           | Customer live GPS tracking on app           | Seller advance payout / BNPL option                 |

## 7. Risks & Mitigations

The following risks have been identified for the implementation programme:

- Seller resistance to stock API mandate — Mitigation: Phase rollout starting with top-500 sellers; offer tech support and settlement incentives for early adopters
- Logistics partner integration complexity — Mitigation: Use an integration middleware layer (iPaaS) rather than point-to-point integrations; pilot with top 2 carriers first
- DE app relaunch causing field disruption — Mitigation: Parallel run period of 4 weeks; dedicated field support team; phased city-by-city rollout
- Data quality issues in existing OMS/WMS — Mitigation: Run a data cleansing sprint in parallel with Phase 1; establish data governance committee
- Change management fatigue across ops teams — Mitigation: Appoint change champions in each hub; link KPI bonuses to new process adherence

## 8. Recommended Next Steps

- Week 1–2: Circulate this report to CTO, VP Operations, VP Logistics, and Seller Management Head for sign-off
  - Week 2–3: Conduct a prioritisation workshop to confirm Phase 1 scope, assign workstream owners, and set OKRs
  - Week 3–4: Issue RFPs for logistics orchestration middleware and WMS upgrade vendors
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- Month 2: Kick off seller API pilot with top 500 sellers; begin DE app redesign sprint
  - Month 3: Deploy payment retry engine and omnichannel notification engine (both relatively low-risk, high-impact)
  - Ongoing: Establish a fortnightly fulfillment performance review cadence using the KPI dashboard built in Phase 3

## 9. Appendix: Stakeholders & Glossary

### 9.1 Key Stakeholders

|                                  |                                                      |
|----------------------------------|------------------------------------------------------|
| <b>Chief Technology Officer</b>  | Technology strategy, system integration approvals    |
| <b>VP Operations</b>             | Warehouse, hub, and last-mile execution              |
| <b>VP Logistics</b>              | Carrier partnerships, first-mile and last-mile SLAs  |
| <b>Seller Management Head</b>    | Seller onboarding, SLA enforcement, incentive design |
| <b>Customer Care Head</b>        | WISMO reduction, return experience                   |
| <b>Finance / CFO</b>             | Settlement cycle, working capital, RTO cost          |
| <b>Business Analyst (Author)</b> | Process documentation, gap analysis, solution design |

### 9.2 Glossary

|                 |                                                                              |
|-----------------|------------------------------------------------------------------------------|
| <b>OMS</b>      | Order Management System                                                      |
| <b>WMS</b>      | Warehouse Management System                                                  |
| <b>DE</b>       | Delivery Executive                                                           |
| <b>RTO</b>      | Return to Origin — package returned to seller after failed delivery attempts |
| <b>WISMO</b>    | Where Is My Order — customer enquiry type                                    |
| <b>SLA</b>      | Service Level Agreement                                                      |
| <b>Tier 2/3</b> | Indian cities outside the top 8 metros                                       |
| <b>iPaaS</b>    | Integration Platform as a Service                                            |
| <b>T+N</b>      | Settlement N business days after delivery confirmation                       |
| <b>ERP</b>      | Enterprise Resource Planning system (used by larger sellers)                 |

— End of Report —

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